

Answer **all** questions.

1. Ffion is investigating a neutralisation reaction between sulfuric acid and sodium hydroxide.

- (a) Use the correct words from the box to complete the **word** equation for this reaction below. [1]

sodium chloride	sodium nitrate	sodium sulfate
-----------------	----------------	----------------

sulfuric acid + sodium hydroxide $\longrightarrow$ + water

- (b) In the experiment she monitors the change of pH using both a pH meter and an indicator.

The chart below shows the colour of phenolphthalein indicator at different pH.

pH scale	1	2	3	4	5	6	7	8	9	10	11	12	13	14
phenolphthalein	colourless								pale pink	pink				

- (i) The table below shows Ffion's results. Complete the table. [2]

Volume of sodium hydroxide added (cm ³)	pH meter reading	Phenolphthalein indicator colour
0	1.0	colourless
5	1.0	colourless
10	1.1	colourless
15	1.2	colourless
20	2.0	
25	9.0	pale pink
30	13.0	
35	14.0	pink
40	14.0	pink



- (ii) Use the information in the table to tick (✓) the boxes next to the **three** correct statements. [3]

Before any sodium hydroxide is added to the acid the phenolphthalein turns green.

☐

The pH is lowest at the start of the experiment.

☐

When 10 cm³ of sodium hydroxide has been added the solution is neutral.

☐

As sodium hydroxide is added the pH rises.

☐

The pH is highest at the end of the experiment.

☐

When 40 cm³ of sodium hydroxide has been added the solution is acidic.

☐

- (c) Sodium hydroxide (NaOH) contains the three elements sodium, hydrogen and oxygen.

- (i) Select the correct numbers from the box to complete the table below. [3]

1	2	8	11	12	16
---	---	---	----	----	----

Element	Number of protons	Number of neutrons	Number of electrons	Mass number
sodium	11	12		23
oxygen	8		8	16
hydrogen	1	0	1	

- (ii) Sodium metal is found in Group 1 of the Periodic Table. Use the Periodic Table on page 20 to name **one** other metal found in this group. [1]

.....

